

Module 08: Planning in Middle School

Learning Objectives

1. Define **Planning** within the context of Middle School.
2. Analyze how Planning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Middle School to the formal definition of Planning.

Introduction

This module explores **Planning**. In the **Middle School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Planning is a critical component of the 8-part Active Inference spine, bridging the gap between Communication and Systems.

Key Concepts

1. Planning as a Markov Blanket Boundary

How does Planning define the boundary between the agent and the environment?

2. Generative Models of Planning

What parameters involved in Planning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Planning drive the perception-action loop?

Applications

In Middle School, we see Planning manifest in: * **Specific Example 1:** Before writing a single line of code for a game, experienced programmers plan by sketching out the major pieces --

player movement, scoring, enemies, levels -- and how they connect. This is like your brain simulating the finished project in advance, predicting where the tricky parts will be, and deciding what to build first. Without that plan, you end up rewriting the same code three times because the pieces do not fit together. * **Specific Example 2:** A GPS navigation app does planning in real time. It builds a model of every possible route from your location to the destination, predicts how long each will take based on traffic data, and selects the path with the lowest predicted travel time. When it detects unexpected traffic (a prediction error), it re-plans by simulating alternative routes and switching you to a better one -- that is active inference planning running on a computer.

Conclusion

Understanding Planning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.